

Linear Statistical Models — Homework 2

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References to lecture notes. [GM] *Gauss–Markov Framework for Linear Models*; [SLR] *Overview of Simple Linear Regression*; [PMI] *Projection and Matrix Inversion*. Throughout, $\chi_k^2(\lambda)$ denotes the noncentral chi-square law with k degrees of freedom and noncentrality λ : if $Z_i \sim N(a_i, 1)$ are independent, then $\sum_{i=1}^k Z_i^2 \sim \chi_k^2(\frac{1}{2} \sum_i a_i^2)$; and $F_{m,d}(\lambda)$ is the law of $\frac{U/m}{V/d}$ with $U \sim \chi_m^2(\lambda) \perp V \sim \chi_d^2$.

Problem 1

Consider the linear model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, where $\mathbf{X}$ is $n \times p$ with $\text{rank}(\mathbf{X}) \leq p$. Define $\boldsymbol{\mu} = \mathbb{E}(\mathbf{Y})$ and $\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\mu}$.

- (a) Show that the least squares estimator of $\boldsymbol{\mu}$ is $\hat{\boldsymbol{\mu}} = \mathbf{P}_\mathbf{X} \mathbf{Y}$, where $\mathbf{P}_\mathbf{X}$ is the orthogonal projection onto the column space of $\mathbf{X}$.
- (b) Show that $\boldsymbol{\theta}$ is estimable and its BLUE is $\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{Y}$.
- (c) Show that if $\mathbf{a}^\top \boldsymbol{\beta}$ is estimable, then its BLUE is $\mathbf{w}^\top \hat{\boldsymbol{\theta}}$ for some $\mathbf{w} \in \mathbb{R}^p$.

Let $\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ be the orthogonal projector onto $\mathcal{C}(\mathbf{X})$ (unique, independent of the g-inverse, [PMI]).

(a) $\hat{\boldsymbol{\mu}} = \mathbf{P}_\mathbf{X} \mathbf{Y}$

Since $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ ranges over $\mathcal{C}(\mathbf{X})$ as $\boldsymbol{\beta}$ varies, the least-squares estimate of $\boldsymbol{\mu}$ minimises $\|\mathbf{Y} - \mathbf{v}\|^2$ over $\mathbf{v} \in \mathcal{C}(\mathbf{X})$. For any such $\mathbf{v}$, $\mathbf{Y} - \mathbf{v} = (\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y} + (\mathbf{P}_\mathbf{X}\mathbf{Y} - \mathbf{v})$, where $(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y} \in \mathcal{C}(\mathbf{X})^\perp$ and $\mathbf{P}_\mathbf{X}\mathbf{Y} - \mathbf{v} \in \mathcal{C}(\mathbf{X})$ are orthogonal, so by Pythagoras

$$\|\mathbf{Y} - \mathbf{v}\|^2 = \|(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y}\|^2 + \|\mathbf{P}_\mathbf{X}\mathbf{Y} - \mathbf{v}\|^2 \geq \|(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y}\|^2,$$

with equality iff $\mathbf{v} = \mathbf{P}_\mathbf{X}\mathbf{Y}$. As $\mathbf{P}_\mathbf{X}\mathbf{Y} \in \mathcal{C}(\mathbf{X})$, the minimiser is $\hat{\boldsymbol{\mu}} = \mathbf{P}_\mathbf{X}\mathbf{Y}$. ■

(b) $\boldsymbol{\theta}$ estimable, $\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{Y}$

$\boldsymbol{\theta} = \mathbf{X}^\top \boldsymbol{\mu} = \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}$, so its i -th component is $\theta_i = \mathbf{a}_i^\top \boldsymbol{\beta}$ with $\mathbf{a}_i = (i\text{-th column of } \mathbf{X}^\top \mathbf{X}) \in \mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{R}(\mathbf{X})$ (equal column spaces since $\text{rank}(\mathbf{X}^\top \mathbf{X}) = \text{rank}(\mathbf{X})$, [PMI]). By the estimability criterion [GM, Lemma 1] each θ_i , hence $\boldsymbol{\theta}$, is estimable. By the Gauss–Markov theorem [GM, Thm 1] the unique BLUE of $\theta_i = \mathbf{a}_i^\top \boldsymbol{\beta}$ is $\mathbf{a}_i^\top \hat{\boldsymbol{\beta}}$, i.e. $\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}$ for any solution $\hat{\boldsymbol{\beta}}$ of the normal equations $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y}$. Therefore

$$\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{Y},$$

a value unique even when $\hat{\boldsymbol{\beta}}$ is not. (Equivalently $\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \hat{\boldsymbol{\mu}} = \mathbf{X}^\top \mathbf{P}_\mathbf{X} \mathbf{Y} = \mathbf{X}^\top \mathbf{Y}$, using $\mathbf{X}^\top \mathbf{P}_\mathbf{X} = \mathbf{X}^\top$.) ■

(c) BLUE of $\mathbf{a}^\top \boldsymbol{\beta}$ is $\mathbf{w}^\top \hat{\boldsymbol{\theta}}$

Estimability gives $\mathbf{a} \in \mathcal{R}(\mathbf{X}) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ [GM, Lemma 1; PMI], so $\mathbf{a} = \mathbf{X}^\top \mathbf{X} \mathbf{w}$ for some $\mathbf{w} \in \mathbb{R}^p$. Then $\mathbf{a}^\top \boldsymbol{\beta} = \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{w}^\top \boldsymbol{\theta}$, and by the Gauss–Markov theorem the BLUE of $\mathbf{a}^\top \boldsymbol{\beta}$ is

$$\mathbf{a}^\top \hat{\boldsymbol{\beta}} = \mathbf{w}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{w}^\top \mathbf{X}^\top \mathbf{Y} = \mathbf{w}^\top \hat{\boldsymbol{\theta}},$$

with $\mathbf{w}$ any solution of $\mathbf{X}^\top \mathbf{X} \mathbf{w} = \mathbf{a}$. ■

Problem 2

Consider $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\beta} \in \mathbb{R}^p$, $\text{rank}(\mathbf{X}) = p$, $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, and LSE $\hat{\boldsymbol{\beta}}$. Let $\mathbf{X}_{\cdot j}$ be the j -th column of $\mathbf{X}$ and $\mathbf{X}_{-j}$ the $n \times (p-1)$ submatrix with the j -th column removed.

- (a) Show that $\text{Var}(\hat{\beta}_j) = \sigma^2 / (\mathbf{X}_{\cdot j}^\top (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j})$, where $\mathbf{P}_{-j}$ is the orthogonal projector onto $\mathcal{C}(\mathbf{X}_{-j})$.
- (b) For the one-variable submodel $\mathbf{Y} = \mathbf{X}_{\cdot j} \beta_j + \boldsymbol{\varepsilon}$, find the distribution of the LSE $\tilde{\beta}_j$ (i) if this submodel is true, and (ii) if instead the full model is true.
- (c) Argue why the submodel is preferable for estimating β_j when it is correct; and state a disadvantage of the submodel when it is incorrect.
- (d) When do the distributions of $\hat{\beta}_j$ (full model) and $\tilde{\beta}_j$ (submodel) coincide?

Here $\text{Var}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$.

(a) $\text{Var}(\hat{\beta}_j) = \sigma^2 / (\mathbf{X}_{\cdot j}^\top (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j})$

Relabelling columns, take $j = p$ (a permutation of columns permutes the entries of $\hat{\boldsymbol{\beta}}$ and the rows/columns of $(\mathbf{X}^\top \mathbf{X})^{-1}$ identically, leaving $\text{Var}(\hat{\beta}_j)$ unchanged). Partition $\mathbf{X} = [\mathbf{X}_{-p} : \mathbf{X}_p]$, so

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} A & b \\ b^\top & d \end{pmatrix}, \quad A = \mathbf{X}_{-p}^\top \mathbf{X}_{-p} \text{ (nonsingular)}, \quad b = \mathbf{X}_{-p}^\top \mathbf{X}_p, \quad d = \mathbf{X}_p^\top \mathbf{X}_p.$$

Let the last column of $(\mathbf{X}^\top \mathbf{X})^{-1}$ be $\begin{pmatrix} g \\ c \end{pmatrix}$; then $\mathbf{X}^\top \mathbf{X} \begin{pmatrix} g \\ c \end{pmatrix} = \mathbf{e}_p = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ gives $Ag + bc = \mathbf{0}$ and $b^\top g + dc = 1$, whence $g = -cA^{-1}b$ and $c(d - b^\top A^{-1}b) = 1$. Thus the (p, p) entry of $(\mathbf{X}^\top \mathbf{X})^{-1}$ is $c = (d - b^\top A^{-1}b)^{-1}$, and with $\mathbf{P}_{-p} = \mathbf{X}_{-p}(\mathbf{X}_{-p}^\top \mathbf{X}_{-p})^{-1} \mathbf{X}_{-p}^\top$,

$$d - b^\top A^{-1}b = \mathbf{X}_p^\top \mathbf{X}_p - \mathbf{X}_p^\top \mathbf{X}_{-p} (\mathbf{X}_{-p}^\top \mathbf{X}_{-p})^{-1} \mathbf{X}_{-p}^\top \mathbf{X}_p = \mathbf{X}_p^\top (\mathbf{I} - \mathbf{P}_{-p}) \mathbf{X}_p.$$

Hence $\text{Var}(\hat{\beta}_p) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{pp} = \sigma^2 / (\mathbf{X}_p^\top (\mathbf{I} - \mathbf{P}_{-p}) \mathbf{X}_p)$. ■

(b) Distribution of $\tilde{\beta}_j$

$\tilde{\beta}_j = \frac{\mathbf{X}_{\cdot j}^\top \mathbf{Y}}{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}}$ is a linear function of the normal vector $\mathbf{Y}$, hence normal; its variance is $\frac{\mathbf{X}_{\cdot j}^\top (\sigma^2 \mathbf{I}) \mathbf{X}_{\cdot j}}{(\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j})^2} = \frac{\sigma^2}{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}}$ in either case.

- (i) *Submodel true:* $\mathbb{E}(\mathbf{Y}) = \mathbf{X}_{\cdot j}\beta_j \Rightarrow \mathbb{E}(\tilde{\beta}_j) = \beta_j$, so $\tilde{\beta}_j \sim N(\beta_j, \sigma^2/\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j})$.
(ii) *Full model true:* $\mathbb{E}(\mathbf{Y}) = \mathbf{X}_{\cdot j}\beta_j + \mathbf{X}_{-j}\beta_{-j}$, so

$$\tilde{\beta}_j \sim N\left(\beta_j + \frac{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{-j}\beta_{-j}}{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}}, \frac{\sigma^2}{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}}\right),$$

same variance but biased by $\mathbf{X}_{\cdot j}^\top \mathbf{X}_{-j}\beta_{-j}/\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}$ (omitted-variable bias).

(c) Efficiency vs. bias

$\mathbf{P}_{-j} \succeq 0$, so $\mathbf{X}_{\cdot j}^\top \mathbf{P}_{-j} \mathbf{X}_{\cdot j} \geq 0$ and hence $\mathbf{X}_{\cdot j}^\top (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} \leq \mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}$, giving

$$\text{Var}(\tilde{\beta}_j) = \frac{\sigma^2}{\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}} \leq \frac{\sigma^2}{\mathbf{X}_{\cdot j}^\top (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j}} = \text{Var}(\hat{\beta}_j).$$

So if the submodel is correct, $\tilde{\beta}_j$ is unbiased and at least as efficient as $\hat{\beta}_j$ (strictly, unless $\mathbf{P}_{-j} \mathbf{X}_{\cdot j} = \mathbf{0}$) — hence preferable. If the submodel is incorrect (full model holds), $\tilde{\beta}_j$ carries the nonvanishing, unknown bias of (b)(ii); tests and confidence intervals based on it are centred at the wrong value and are therefore invalid.

(d) When the laws coincide

If $\mathbf{X}_{\cdot j} \perp \mathcal{C}(\mathbf{X}_{-j})$, i.e. $\mathbf{X}_{\cdot j}^\top \mathbf{X}_{-j} = \mathbf{0}$, then $\mathbf{P}_{-j} \mathbf{X}_{\cdot j} = \mathbf{0}$, so $\mathbf{X}_{\cdot j}^\top (\mathbf{I} - \mathbf{P}_{-j}) \mathbf{X}_{\cdot j} = \mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j}$ and the bias vanishes. Both estimators are then $N(\beta_j, \sigma^2/\mathbf{X}_{\cdot j}^\top \mathbf{X}_{\cdot j})$, so $\hat{\beta}_j \stackrel{d}{=} \tilde{\beta}_j$ — exactly when column $\mathbf{X}_{\cdot j}$ is orthogonal to all other columns. ■

Problem 3

Consider $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_{p-1} x_{i,p-1} + \varepsilon_i$, $\varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$.

(a) Show that $R^2 = \frac{\tilde{\beta}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \tilde{\beta}}{\sum_{i=1}^n (Y_i - \bar{Y})^2}$, where $\tilde{\mathbf{X}}$ is the $n \times (p-1)$ matrix with (i, j) -entry $x_{ij} - \bar{x}_{\cdot j}$,

$\bar{x}_{\cdot j} = \frac{1}{n} \sum_i x_{ij}$, and $\tilde{\beta}$ is the BLUE of $(\beta_1, \dots, \beta_{p-1})^\top$.

(b) Find the distribution of $R^2/(1 - R^2)$, assuming the model is true.

Write the model as $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$, $\mathbf{X} = [\mathbf{1} : \cdots]$ ($n \times p$, rank p , first column $\mathbf{1}$). Let $\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ and $\mathbf{P}_1 = \frac{1}{n} \mathbf{1}\mathbf{1}^\top$ (projector onto $\mathcal{C}(\mathbf{1})$). Since $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathbf{P}_\mathbf{X} \mathbf{P}_1 = \mathbf{P}_1$.

(a) $R^2 = \tilde{\beta}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \tilde{\beta} / \sum_i (Y_i - \bar{Y})^2$

With $\hat{\beta} = (\hat{\beta}_0, \dots, \hat{\beta}_{p-1})^\top$ the LSE and $\hat{\mathbf{Y}} = \mathbf{P}_\mathbf{X} \mathbf{Y}$,

$$\text{SSTO} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{Y}, \quad \text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_\mathbf{X}) \mathbf{Y}, \quad \text{SSR} = \text{SSTO} - \text{SSE} = \mathbf{Y}^\top (\mathbf{P}_\mathbf{X} - \mathbf{P}_1) \mathbf{Y}.$$

As $\mathbf{1} \in \mathcal{C}(\mathbf{X})$, $\mathbf{1}^\top(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y} = \mathbf{0}$, i.e. $\mathbf{1}^\top\mathbf{Y} = \mathbf{1}^\top\mathbf{X}\hat{\boldsymbol{\beta}}$, so $\mathbf{P}_1\mathbf{Y} = \mathbf{P}_1\mathbf{X}\hat{\boldsymbol{\beta}}$. Using symmetry/idempotence of the projectors,

$$\text{SSR} = \|\mathbf{P}_\mathbf{X}\mathbf{Y}\|^2 - \|\mathbf{P}_1\mathbf{Y}\|^2 = \|\mathbf{X}\hat{\boldsymbol{\beta}}\|^2 - \|\mathbf{P}_1\mathbf{X}\hat{\boldsymbol{\beta}}\|^2 = \hat{\boldsymbol{\beta}}^\top\mathbf{X}^\top(\mathbf{I} - \mathbf{P}_1)\mathbf{X}\hat{\boldsymbol{\beta}} = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{X}\hat{\boldsymbol{\beta}}\|^2.$$

Now $(\mathbf{I} - \mathbf{P}_1)\mathbf{X}$ centres each column: its first column is $(\mathbf{I} - \mathbf{P}_1)\mathbf{1} = \mathbf{0}$ and its $(j+1)$ -th column has i -th entry $x_{ij} - \bar{x}_{.j}$, i.e. $(\mathbf{I} - \mathbf{P}_1)\mathbf{X} = [\mathbf{0} : \tilde{\mathbf{X}}]$. Hence $(\mathbf{I} - \mathbf{P}_1)\mathbf{X}\hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}\tilde{\boldsymbol{\beta}}$ with $\tilde{\boldsymbol{\beta}} = (\hat{\beta}_1, \dots, \hat{\beta}_{p-1})^\top$, and

$$\text{SSR} = \tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \tilde{\boldsymbol{\beta}}, \quad R^2 = \frac{\text{SSR}}{\text{SSTO}} = \frac{\tilde{\boldsymbol{\beta}}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \tilde{\boldsymbol{\beta}}}{\sum_{i=1}^n (Y_i - \bar{Y})^2}. \quad \blacksquare$$

(b) Distribution of $R^2/(1 - R^2)$

Since $R^2 = \text{SSR}/\text{SSTO}$ and $1 - R^2 = \text{SSE}/\text{SSTO}$, $\frac{R^2}{1 - R^2} = \frac{\text{SSR}}{\text{SSE}} = \frac{\mathbf{Y}^\top(\mathbf{P}_\mathbf{X} - \mathbf{P}_1)\mathbf{Y}}{\mathbf{Y}^\top(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{Y}}$. Both $\mathbf{P}_\mathbf{X} - \mathbf{P}_1$ and $\mathbf{I} - \mathbf{P}_\mathbf{X}$ are orthogonal projectors (symmetric idempotent; e.g. $(\mathbf{P}_\mathbf{X} - \mathbf{P}_1)^2 = \mathbf{P}_\mathbf{X} - \mathbf{P}_1$ using $\mathbf{P}_\mathbf{X}\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_\mathbf{X} = \mathbf{P}_1$) of ranks $p - 1$ and $n - p$, with product $(\mathbf{P}_\mathbf{X} - \mathbf{P}_1)(\mathbf{I} - \mathbf{P}_\mathbf{X}) = \mathbf{0}$.

Write a rank- m orthogonal projector as $\mathbf{V}\mathbf{V}^\top$ with $\mathbf{V}$ ($n \times m$) having orthonormal columns spanning its range. So $\mathbf{P}_\mathbf{X} - \mathbf{P}_1 = \mathbf{V}_1\mathbf{V}_1^\top$ and $\mathbf{I} - \mathbf{P}_\mathbf{X} = \mathbf{V}_2\mathbf{V}_2^\top$ with $\mathbf{V}_1^\top\mathbf{V}_1 = \mathbf{I}_{p-1}$, $\mathbf{V}_2^\top\mathbf{V}_2 = \mathbf{I}_{n-p}$; the product being $\mathbf{0}$ forces $\mathbf{V}_1^\top\mathbf{V}_2 = \mathbf{0}$. Then $\text{SSR} = \|\mathbf{V}_1^\top\mathbf{Y}\|^2$, $\text{SSE} = \|\mathbf{V}_2^\top\mathbf{Y}\|^2$, and with $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ the vectors $\mathbf{V}_1^\top\mathbf{Y}, \mathbf{V}_2^\top\mathbf{Y}$ are jointly normal with $\text{Cov}(\mathbf{V}_1^\top\mathbf{Y}, \mathbf{V}_2^\top\mathbf{Y}) = \sigma^2\mathbf{V}_1^\top\mathbf{V}_2 = \mathbf{0}$, hence independent; therefore $\text{SSR} \perp \text{SSE}$. Moreover $\frac{1}{\sigma}\mathbf{V}_1^\top\mathbf{Y} \sim N_{p-1}(\frac{1}{\sigma}\mathbf{V}_1^\top\mathbf{X}\boldsymbol{\beta}, \mathbf{I})$ and $(\mathbf{I} - \mathbf{P}_\mathbf{X})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$, so

$$\frac{\text{SSR}}{\sigma^2} \sim \chi_{p-1}^2(\lambda), \quad \lambda = \frac{(\mathbf{X}\boldsymbol{\beta})^\top(\mathbf{P}_\mathbf{X} - \mathbf{P}_1)(\mathbf{X}\boldsymbol{\beta})}{2\sigma^2} = \frac{\boldsymbol{\beta}_{(1)}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \boldsymbol{\beta}_{(1)}}{2\sigma^2}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi_{n-p}^2,$$

where $\boldsymbol{\beta}_{(1)} = (\beta_1, \dots, \beta_{p-1})^\top$ (true values) and we used $(\mathbf{P}_\mathbf{X} - \mathbf{P}_1)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}\boldsymbol{\beta} = \tilde{\mathbf{X}}\boldsymbol{\beta}_{(1)}$ as in (a). Hence

$$\frac{n-p}{p-1} \cdot \frac{R^2}{1-R^2} = \frac{\text{SSR}/(p-1)}{\text{SSE}/(n-p)} \sim F_{p-1, n-p}(\lambda), \quad \lambda = \frac{\boldsymbol{\beta}_{(1)}^\top \tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \boldsymbol{\beta}_{(1)}}{2\sigma^2}.$$

Under $H_0 : \beta_1 = \dots = \beta_{p-1} = 0$, $\lambda = 0$ and the law is central $F_{p-1, n-p}$. ■

Problem 4

Let $\mathbf{A}$ be $n \times n$ positive semidefinite with $\text{rank}(\mathbf{A}) = r$ and $\text{rank}(\mathbf{I}_n - \mathbf{A}) = n - r$, $1 \leq r < n$. Prove that $\mathbf{A}$ is an orthogonal projection matrix. Further, if $\mathbf{Y} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$, show (without Fisher-Cochran) that $U_1 := \mathbf{Y}^\top\mathbf{A}\mathbf{Y}$ and $U_2 := \mathbf{Y}^\top\mathbf{Y} - \mathbf{Y}^\top\mathbf{A}\mathbf{Y}$ are independent, with $U_1 \sim \chi_r^2(\delta_1/2)$ and $U_2 \sim \chi_{n-r}^2(\delta_2/2)$, where $\delta_1 = \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu}$ and $\delta_2 = \boldsymbol{\mu}^\top\boldsymbol{\mu} - \boldsymbol{\mu}^\top\mathbf{A}\boldsymbol{\mu}$.

$\mathbf{A}$ is an orthogonal projector

Being symmetric positive semidefinite, $\mathbf{A}$ has a spectral decomposition $\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$ with $\mathbf{U}$ orthogonal and $\mathbf{D}$ diagonal; its diagonal entries are the eigenvalues of $\mathbf{A}$, of which exactly r are positive

(as $\text{rank } \mathbf{A} = r$). Ordering them,

$$\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_r, 0, \dots, 0), \quad \lambda_i > 0,$$

$$\mathbf{I}_n - \mathbf{D} = \text{diag}(1 - \lambda_1, \dots, 1 - \lambda_r, \underbrace{1, \dots, 1}_{n-r}).$$

Then $\mathbf{I}_n - \mathbf{A} = \mathbf{U}(\mathbf{I}_n - \mathbf{D})\mathbf{U}^\top$; its last $n-r$ diagonal entries equal $1 \neq 0$, and since $\text{rank}(\mathbf{I}_n - \mathbf{A}) = n-r$ the remaining r vanish, i.e. $\lambda_i = 1$ for all i . Hence $\mathbf{D}^2 = \mathbf{D}$, so $\mathbf{A}^2 = \mathbf{U}\mathbf{D}^2\mathbf{U}^\top = \mathbf{A}$; being symmetric and idempotent, $\mathbf{A}$ is an orthogonal projector of rank r (and $\mathbf{I}_n - \mathbf{A}$ one of rank $n-r$). ■

Independence and distributions

With $\mathbf{D} = \text{diag}(1, \dots, 1, 0, \dots, 0)$ (r ones), set $\mathbf{Z} = \mathbf{U}^\top \mathbf{Y}$. As $\mathbf{U}$ is orthogonal and $\mathbf{Y} \sim N(\boldsymbol{\mu}, \mathbf{I}_n)$, we have $\mathbf{Z} \sim N(\mathbf{U}^\top \boldsymbol{\mu}, \mathbf{I}_n)$, so $Z_1, \dots, Z_n$ are independent with $Z_i \sim N((\mathbf{U}^\top \boldsymbol{\mu})_i, 1)$. Then

$$U_1 = \mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Z}^\top \mathbf{D} \mathbf{Z} = \sum_{i=1}^r Z_i^2, \quad U_2 = \mathbf{Y}^\top (\mathbf{I}_n - \mathbf{A}) \mathbf{Y} = \sum_{i=r+1}^n Z_i^2.$$

By the noncentral- χ^2 convention,

$$U_1 \sim \chi_r^2(\delta_1/2), \quad \delta_1 = \sum_{i=1}^r (\mathbf{U}^\top \boldsymbol{\mu})_i^2 = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu},$$

$$U_2 \sim \chi_{n-r}^2(\delta_2/2), \quad \delta_2 = \sum_{i=r+1}^n (\mathbf{U}^\top \boldsymbol{\mu})_i^2 = \boldsymbol{\mu}^\top (\mathbf{I}_n - \mathbf{A}) \boldsymbol{\mu} = \boldsymbol{\mu}^\top \boldsymbol{\mu} - \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

Finally U_1 is a function of $(Z_1, \dots, Z_r)$ only and U_2 of $(Z_{r+1}, \dots, Z_n)$ only; since $Z_1, \dots, Z_n$ are mutually independent, $U_1 \perp U_2$. (No Fisher–Cochran theorem was used.) ■

Problem 5

Consider $Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \varepsilon_i$, $i = 1, \dots, 8$, with ε_i uncorrelated, mean 0, variance σ^2 . Estimate $\beta_1, \beta_2, \sigma^2$ from:

i	1	2	3	4	5	6	7	8
Y_i	82	79	74	83	80	81	84	81
x_{i1}	10	9	9	11	11	10	10	12
x_{i2}	15	14	13	15	14	14	16	13

$\mathbf{X} = [\mathbf{x}_{\cdot 1} : \mathbf{x}_{\cdot 2}]$ is 8×2 of rank 2, so $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$. From the data,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \sum_i x_{i1}^2 & \sum_i x_{i1} x_{i2} \\ \sum_i x_{i1} x_{i2} & \sum_i x_{i2}^2 \end{pmatrix} = \begin{pmatrix} 848 & 1168 \\ 1168 & 1632 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{Y} = \begin{pmatrix} \sum_i x_{i1} Y_i \\ \sum_i x_{i2} Y_i \end{pmatrix} = \begin{pmatrix} 6612 \\ 9194 \end{pmatrix},$$

and $\sum_i Y_i^2 = 51908$. $\det(\mathbf{X}^\top \mathbf{X}) = 848 \cdot 1632 - 1168^2 = 19712$, so $(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{19712} \begin{pmatrix} 1632 & -1168 \\ -1168 & 848 \end{pmatrix}$ and

$$\hat{\beta}_1 = \frac{1632 \cdot 6612 - 1168 \cdot 9194}{19712} = \frac{233}{88} \approx 2.6477, \quad \hat{\beta}_2 = \frac{848 \cdot 9194 - 1168 \cdot 6612}{19712} = \frac{329}{88} \approx 3.7386.$$

With residual degrees of freedom $n - \text{rank}(\mathbf{X}) = 8 - 2 = 6$,

$$\text{SSE} = \mathbf{Y}^\top \mathbf{Y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{Y} = 51908 - \left(\frac{233}{88} \cdot 6612 + \frac{329}{88} \cdot 9194 \right) = \frac{1241}{44} \approx 28.205,$$

$$\hat{\sigma}^2 = \frac{\text{SSE}}{n - 2} = \frac{1241}{264} \approx 4.701.$$

(For reference, $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = \text{MSE}(\mathbf{X}^\top \mathbf{X})^{-1}$ gives $s(\hat{\beta}_1) \approx 0.624$, $s(\hat{\beta}_2) \approx 0.450$.) ■

Problem 6

Consider $Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 + \beta_{12} x_{i1} x_{i2} + \varepsilon_i$, $i = 1, \dots, 7$, with predictors:

i	1	2	3	4	5	6	7
x_{i1}	1	1	-1	-1	0	0	0
x_{i2}	1	-1	1	-1	0	0	0

Show that $\beta_0, \beta_1, \beta_2, \beta_{11} + \beta_{22}, \beta_{12}$ are estimable and find nonmatrix algebraic forms for their estimates. Find the MSE and the standard errors of the estimates.

The six columns of $\mathbf{X}$ (order $\beta_0, \beta_1, \beta_2, \beta_{11}, \beta_{22}, \beta_{12}$) are

$$\begin{aligned} \mathbf{c}_0 &= (1, 1, 1, 1, 1, 1)^\top, & \mathbf{c}_1 &= (1, 1, -1, -1, 0, 0)^\top, & \mathbf{c}_2 &= (1, -1, 1, -1, 0, 0)^\top, \\ \mathbf{c}_{11} &= (1, 1, 1, 1, 0, 0)^\top, & \mathbf{c}_{22} &= (1, 1, 1, 1, 0, 0)^\top, & \mathbf{c}_{12} &= (1, -1, -1, 1, 0, 0)^\top. \end{aligned}$$

Estimability

Since $x_{i1}^2 = x_{i2}^2$ at every design point, $\mathbf{c}_{11} = \mathbf{c}_{22}$; thus $\mathbf{X}\mathbf{v} = \mathbf{0}$ for $\mathbf{v} = (0, 0, 0, 1, -1, 0)^\top$, and in fact $\mathcal{N}(\mathbf{X}) = \text{span}\{\mathbf{v}\}$ (the other five columns $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_{11}, \mathbf{c}_{12}$ are independent: $\mathbf{c}_0$ is the only one nonzero on rows 5, 6, 7, and on rows 1–4 the vectors $\mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_{11}, \mathbf{c}_{12}$ are mutually orthogonal ± 1 patterns). So $\text{rank}(\mathbf{X}) = 5$. A function $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ is estimable iff $\boldsymbol{\lambda} \perp \mathcal{N}(\mathbf{X})$ [GM, Lemma 1], i.e. iff $\lambda_{11} = \lambda_{22}$. This holds for $\beta_0, \beta_1, \beta_2, \beta_{11} + \beta_{22}, \beta_{12}$ (so these are estimable), but not for β_{11} or β_{22} alone. ✓

Estimates

Fit the equivalent full-rank model whose design matrix $\mathbf{Z}$ has columns $\mathbf{c}_0, \mathbf{c}_1, \mathbf{c}_2, \mathbf{c}_{11}, \mathbf{c}_{12}$ and coefficient vector $(\beta_0, \beta_1, \beta_2, \gamma, \beta_{12})$, where $\gamma := \beta_{11} + \beta_{22}$ (valid since $\beta_{11} x_{i1}^2 + \beta_{22} x_{i2}^2 = \gamma \mathbf{c}_{11,i}$). All off-diagonal inner products vanish except $\mathbf{c}_0^\top \mathbf{c}_{11} = 4$:

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} 7 & 0 & 0 & 4 & 0 \\ 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 4 & 0 & 0 \\ 4 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 4 \end{pmatrix}, \quad \mathbf{Z}^\top \mathbf{Y} = \begin{pmatrix} \sum_{i=1}^7 Y_i \\ Y_1 + Y_2 - Y_3 - Y_4 \\ Y_1 - Y_2 + Y_3 - Y_4 \\ Y_1 + Y_2 + Y_3 + Y_4 \\ Y_1 - Y_2 - Y_3 + Y_4 \end{pmatrix}.$$

The orthogonal columns give $\hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_{12}$ directly; $\hat{\beta}_0, \hat{\gamma}$ solve $\begin{pmatrix} 7 & 4 \\ 4 & 4 \end{pmatrix} \begin{pmatrix} \hat{\beta}_0 \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} \sum_{i=1}^7 Y_i \\ \sum_{i=1}^4 Y_i \end{pmatrix}$ (determinant 12). Thus, with $\bar{Y}_c = \frac{1}{3}(Y_5 + Y_6 + Y_7)$ and $\bar{Y}_f = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4)$,

$\hat{\beta}_0 = \frac{1}{3}(Y_5 + Y_6 + Y_7) = \bar{Y}_c,$	$\hat{\beta}_1 = \frac{1}{4}(Y_1 + Y_2 - Y_3 - Y_4),$
$\hat{\beta}_2 = \frac{1}{4}(Y_1 - Y_2 + Y_3 - Y_4),$	$\hat{\beta}_{12} = \frac{1}{4}(Y_1 - Y_2 - Y_3 + Y_4),$
$\widehat{\beta_{11} + \beta_{22}} = \frac{1}{4}(Y_1 + Y_2 + Y_3 + Y_4) - \frac{1}{3}(Y_5 + Y_6 + Y_7) = \bar{Y}_f - \bar{Y}_c.$	

Each is unbiased for the stated function; being least-squares estimates of estimable functions, they are the BLUEs [GM, Thm 1]. (There is no BLUE of β_{11} or β_{22} separately, as neither is estimable.)

MSE and standard errors

The four corner points are fitted exactly (the functions $1, x_1, x_2, x_1x_2$ interpolate any four values at the corners), so all residuals come from the three centre points:

$$\text{SSE} = \sum_{i=5}^7 (Y_i - \bar{Y}_c)^2 = \sum_{i=5}^7 Y_i^2 - \frac{1}{3}(Y_5 + Y_6 + Y_7)^2, \quad \text{d.f.} = n - \text{rank}(\mathbf{X}) = 2, \quad \text{MSE} = \frac{1}{2} \sum_{i=5}^7 (Y_i - \bar{Y}_c)^2.$$

Each estimate is a linear form $\mathbf{a}^\top \mathbf{Y}$ with variance $\sigma^2 \|\mathbf{a}\|^2$ (equivalently σ^2 times a diagonal entry of $(\mathbf{Z}^\top \mathbf{Z})^{-1}$, namely $\frac{1}{3}, \frac{1}{4}, \frac{1}{4}, \frac{7}{12}, \frac{1}{4}$ for $\beta_0, \beta_1, \beta_2, \gamma, \beta_{12}$). Estimating σ^2 by MSE,

$$s(\hat{\beta}_0) = \sqrt{\frac{\text{MSE}}{3}}, \quad s(\hat{\beta}_1) = s(\hat{\beta}_2) = s(\hat{\beta}_{12}) = \frac{1}{2}\sqrt{\text{MSE}}, \quad s(\widehat{\beta_{11} + \beta_{22}}) = \sqrt{\frac{7\text{MSE}}{12}}.$$

■